

ECMT676 Econometrics II

Tentative Solutions to Problem Set 2

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1. (4.8) (5/25)

(a) (1/5)

$$\alpha_1 = E\left(\frac{\partial E(y|x_1, x_2)}{\partial x_1}\right) = E(\beta_1 + \beta_3 x_2) = \beta_1 + \beta_3 \mu_2$$

$$\alpha_2 = E\left(\frac{\partial E(y|x_1, x_2)}{\partial x_2}\right) = E(\beta_2 + \beta_3 x_1 + 2\beta_4 x_2) = \beta_2 + \beta_3 \mu_1 + 2\beta_4 \mu_2$$

(b) (1/5) First, we can rewrite β_1 and β_2 as $\beta_1 = \alpha_1 - \beta_3 \mu_2$ and $\beta_2 = \alpha_2 - \beta_3 \mu_1 - 2\beta_4 \mu_2$. Then,

$$E(y|x_1, x_2) = \beta_0 + (\alpha_1 - \beta_3 \mu_2)x_1 + (\alpha_2 + \beta_3 \mu_1 - 2\beta_4 \mu_2)x_2 + \beta_3 x_1 x_2 + \beta_4 x_2^2$$

$$= (\beta_0 - \beta_3 \mu_1 \mu_2 - \beta_4 \mu_2^2) + \alpha_1 x_1 + \alpha_2 x_2 + \beta_3 (x_1 - \mu_1)(x_2 - \mu_2) + \beta_4 (x_2 - \mu_2)^2$$

(c) (1/5) If μ_1 and μ_2 are known, we can run regression y on $x_1, x_2, (x_1 - \mu_1)(x_2 - \mu_2)$, and $(x_2 - \mu_2)^2$. On the other hand, if μ_1 and μ_2 are unknown, we can replace them with $\bar{x}_1$ and $\bar{x}_2$ and run regression y on $x_1, x_2, (x_1 - \bar{x}_1)(x_2 - \bar{x}_2)$, and $(x_2 - \bar{x}_2)^2$.

(d) (2/5)

4.8(d) demeaned

Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
const	5.3921	0.1199	44.9891	0.0000	5.1568	5.6273
educ	0.0838	0.0068	12.2812	0.0000	0.0704	0.0972
exper	0.0224	0.0034	6.5488	0.0000	0.0157	0.0291
educ.exper	0.0045	0.0017	2.6919	0.0072	0.0012	0.0079
exper^2	0.0010	0.0007	1.4658	0.1430	-0.0003	0.0023

4.8(d) not demeaned

Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
const	6.2334	0.3051	20.4280	0.0000	5.6346	6.8323
educ	0.0312	0.0194	1.6117	0.1074	-0.0068	0.0692
exper	-0.0619	0.0327	-1.8912	0.0589	-0.1261	0.0023

educ.exper	0.0045	0.0017	2.6919	0.0072	0.0012	0.0079
exper^2	0.0010	0.0007	1.4658	0.1430	-0.0003	0.0023
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2. (6.12) (6/25)

(a) (4/6) Following the hint, we can rewrite $\sqrt{N}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS})$ as

$$\begin{aligned}\sqrt{N}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}) &= A_1^{-1} \left(N^{-1/2} \sum_{i=1}^n x_i^{*'} u_i \right) - A_2^{-1} \left(N^{-1/2} \sum_{i=1}^n x_i' u_i \right) + op(1) \\ &= \left[N^{-1/2} \sum_{i=1}^n (A_1^{-1} x_i^{*'} - A_2^{-1} x_i') u_i \right] \equiv N^{-1/2} \sum_{i=1}^n w_i\end{aligned}$$

where $A_1 = E(x_i^{*'} x_i^*)$ and $A_2 = E(x_i' x_i)$. Note that $E(w_i) = E[(A_1^{-1} x_i^{*'} - A_2^{-1} x_i') u_i] = 0$, which follows Assumption 2SLS.1, and

$$\begin{aligned}E(w_i w_i') &= E \left\{ [(A_1^{-1} x_i^{*'} - A_2^{-1} x_i') u_i] [u_i (x_i^* A_1^{-1} - x_i A_2^{-1})] \right\} \\ &= \sigma^2 \left\{ A_1^{-1} E(x_i^{*'} x_i^*) A_1^{-1} - A_2^{-1} E(x_i' x_i') A_2^{-1} \right. \\ &\quad \left. - A_1^{-1} E(x_i^{*'} x_i') A_2^{-1} + A_2^{-1} E(x_i' x_i^*) A_1^{-1} \right\} \\ &= \sigma^2 (A_1^{-1} - A_2^{-1}),\end{aligned}$$

where the last equality comes from $E(x_i^{*'} x_i') = E(x_i' x_i^*) = E(x_i x_i')$. Thus, by the central limit theorem, $\sqrt{N}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS}) \xrightarrow{d} N(0, \sigma^2(A_1^{-1} - A_2^{-1}))$.

(b) (2/6) If Assumption 2SLS.3 does not hold, the asymptotic variance of $\sqrt{N}(\hat{\beta}_{2SLS} - \hat{\beta}_{OLS})$ should be

$$\begin{aligned}E(w_i w_i') &= E \left\{ [(A_1^{-1} x_i^{*'} - A_2^{-1} x_i') u_i] [u_i (x_i^* A_1^{-1} - x_i A_2^{-1})] \right\} \\ &= A_1^{-1} E(u_i^2 x_i^{*'} x_i^*) A_1^{-1} - A_2^{-1} E(u_i^2 x_i' x_i') A_2^{-1} \\ &\quad - A_1^{-1} E(u_i^2 x_i^{*'} x_i) A_2^{-1} + A_2^{-1} E(u_i^2 x_i' x_i^*) A_1^{-1}.\end{aligned}$$

Thus, to estimate the variance, $\hat{E}(w_i w_i')$, we can use

$$\begin{aligned}\hat{A}_1 &= n^{-1} \sum x_i^{*'} x_i^* \\ \hat{A}_2 &= n^{-1} \sum x_i' x_i' \\ \hat{E}(u_i^2 x_i^{*'} x_i^*) &= n^{-1} \sum \hat{u}_i^2 x_i^{*'} x_i^* \\ \hat{E}(u_i^2 x_i' x_i') &= n^{-1} \sum \hat{u}_i^2 x_i' x_i' \\ \hat{E}(u_i^2 x_i^{*'} x_i) &= n^{-1} \sum \hat{u}_i^2 x_i^{*'} x_i \\ \hat{E}(u_i^2 x_i' x_i^*) &= n^{-1} \sum \hat{u}_i^2 x_i' x_i^*\end{aligned}$$

where $\hat{u}_i = y_i - x_i' \hat{\beta}_{OLS}$.

3. (10.2) (6/25)

(a) (1/6) θ_2 , δ_2 , and γ can be estimated consistently.

- (b) (2/6) θ_2 measures the growth in wage for men over period. To see this, notice that

$$E(\log wage_{i,t=1} | male) - E(\log wage_{i,t=2} | male) = \theta_2 + (z_{i,t=1} - z_{i,t=2})\gamma + c_i = \theta_2,$$

provided $z_{i,t=1} = z_{i,t=2}$. On the other hand, δ_2 captures the change of the gap between women and men. To see this, notice that

$$E(\log wage_{i_t} | male) - E(\log wage_{i_t} | female) = \delta_1 + \delta_2 d_{2t}.$$

- (c) (1/6) We can show that $\Delta \log(wage_i) = \theta_2 + \Delta z_i \gamma + \delta_2 female_i + \Delta u_i$.
 (d) (2/6) We can test the hypothesis in the following steps

Step 1 Run regression $\Delta \log(wage_i)$ on a constant, Δz_i , and $female_i$.

Step 2 Compute the robust covariance matrix by

$$\hat{V} \equiv \left(\sum x_i x_i' \right)^{-1} \left(\sum \hat{u}_i^2 x_i x_i' \right) \left(\sum x_i x_i' \right)^{-1},$$

where $x_i = [1, \Delta z_i, female_i]'$ and $\hat{u}_i = \Delta \log(wage_i) - \hat{\theta}_2 + \Delta z_i \hat{\gamma} + \hat{\delta}_2 female_i$.

Step 3 Compute the t -statistics,

$$t = \frac{\hat{\delta}_2}{\sqrt{\hat{V}_{33}}}.$$

Step 4 If $1 - 2 \cdot P(|t| > t_{n-k}) < \alpha$, then we reject H_0 .

4. (10.8) (8/25)

- (a) (3/8)

10.8 (a)

Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
const	4.1812	0.1898	22.0309	0.0000	3.8048	4.5577
d78	-0.0547	0.0867	-0.6314	0.5292	-0.2266	0.1172
crlprc_1	-0.0185	0.0047	-3.9592	0.0001	-0.0278	-0.0092
clcprc_2	-0.0174	0.0045	-3.8879	0.0002	-0.0263	-0.0085

10.8 (a) Test of Serial Correlation

Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
const	0.0000	0.0452	0.0000	1.0000	-0.0907	0.0907
vhat_1	0.5740	0.0968	5.9312	0.0000	0.3797	0.7682

- (b) (3/8)

10.8 (b)

Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
d78	0.0857	0.0539	1.5893	0.1183	-0.0226	0.1939
crlprc_1	-0.0040	0.0041	-0.9769	0.3333	-0.0124	0.0043
clcprc_2	-0.0132	0.0046	-2.8733	0.0059	-0.0224	-0.0040

(c) (2/8)

10.8 (c) Testing $\beta_1 = \beta_2$

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-> Do not reject Null Hypothesis: $\beta_1 = \beta_2$

p-value alpha

0.1643 0.0500

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10.8 (c) Parsimonious Model

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Var.	coef	stde	t-stat	p-val	lb of CI	ub of CI
d78	0.0993	0.0544	1.8252	0.0738	-0.0099	0.2086
crlprc_1+crlprc_2	-0.0083	0.0028	-2.9506	0.0048	-0.0140	-0.0027

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